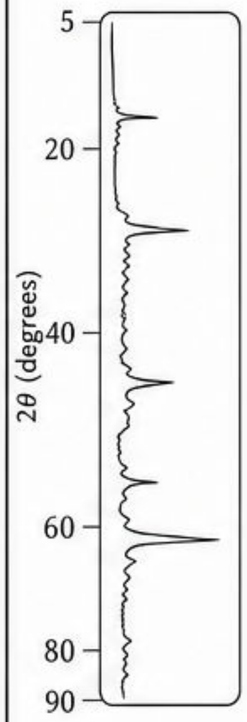
INPUT

PXRD $x \in \mathbb{R}^{8500}$



$2\theta \in [5°, 90°]$

shape: (B, 8500)

BN

Multi-resolution Stem

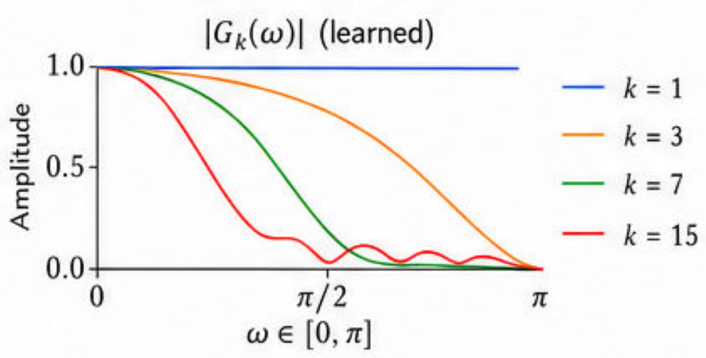
4 parallel Conv1d branches, stride = 2

$k = 1$	$k = 3$	$k = 7$	$k = 15$
level-0 detail (identity)	level-1 (sharp peaks)	level-2 (standard peaks)	level-3 (broadened peaks)
Conv1d(k=1,s=2)	Conv1d(k=3,s=2)	Conv1d(k=7,s=2)	Conv1d(k=15,s=2)
$D/4$	$D/4$	$D/4$	$D/4$

Concatenate → BN + GELU + Conv1d(k=3) + BN + GELU

$h \in \mathbb{R}^{B \times 4251 \times 256}$

learned filter spectra of the stem branches



$h \in \mathbb{R}^{B \times 4251 \times 256}$

soft multi-resolution tiling

MRA-like learned frame

$z \in \mathbb{R}^{B \times 4251 \times 256}$

4-way switch — only this block varies

CNN11-NoPool

8× residual Conv1d

$\text{Conv}_3 \rightarrow \text{BN} \rightarrow \text{GELU} \rightarrow \text{Dropout}$

RF ≈ 33

local | parallel

$O(L \cdot D^2)$

Mamba-2

6× SSD block

LN → (u, Δ)

causal Conv1d(k=4) + SiLU

selective scan

$d_{state} = 16$

recurrent | selective

$O(L \cdot D \cdot d_{state})$

(not run in v4)

GLA

8× gated linear attention

LN → Q, K, V

$\phi = ELU + 1$

gate $G = \sigma(W_g x)$

4× FFN

parallel | global

$O(L \cdot D^2)$

RetNet (chunkwise)

8× retention block

$\gamma = 0.96$

4× FFN

recurrent | chunkwise

$O(L \cdot D^2)$ per chunk

$z \in \mathbb{R}^{B \times 4251 \times 256}$

Pool / Heads / Loss

Mean-pool over L

$z \in \mathbb{R}^{B \times 256}$

Linear D → 230 (SG)

$p_{SG} \in \Delta^{230}$

Linear D → 7 (CS), $\lambda = 0.3$

(disabled in v4)

$p_{CS} \in \Delta^{7}$

$\mathcal{L} = \text{CE}(p_{SG}, y_{SG}) + 0.3 \cdot \text{CE}(p_{CS}, y_{CS})$

Same stem and heads across all runs; only the encoder block is swapped.